



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

CASE STUDY

AI Ethics for AIML Engineers 22AIM821	
Name:	Pavan Aaditya B
USN:	1NH22AI114
Semester:	8 th Semester
Section:	B
Submitted on:	28/04/2026
Submitted to:	Prof. Manoj Kumar



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AI-Based Crime Prediction: A Research-Oriented Case Study Analysis

Introduction

Artificial intelligence has transformed numerous sectors, and law enforcement is no exception. AI-based crime prediction, often termed predictive policing, leverages machine learning, deep learning, and statistical modelling to forecast the likelihood, location, timing, and sometimes perpetrators or victims of criminal activity. These systems analyse vast datasets that include historical crime records, socioeconomic indicators, environmental factors such as weather and time of day, and sometimes real-time inputs like sensor data or CCTV feeds. The goal lies in identifying patterns and allocating police resources proactively. The core premise draws from criminological theories such as routine activity theory and near-repeat victimization. These theories hold that crime exhibits spatiotemporal clustering rather than occurring randomly. Early systems relied on simpler statistical methods such as hot-spot mapping via kernel density estimation. Modern approaches incorporate supervised learning methods like random forests and gradient boosting, neural networks including CNNs and LSTMs for spatiotemporal data, and ensemble methods. Research on urban crime prediction in Indian metropolitan cities has explored deep learning models for real-time risk assessment. AI crime prediction offers potential efficiency gains amid resource constraints in Indian policing. At the same time, these systems raise questions about accuracy, bias, ethics, transparency, and civil liberties. This long-format case study examines the technology foundations, implementations in the Indian context, empirical outcomes, challenges, and future directions.

Technical Foundations and Methodologies

AI crime prediction typically falls into two categories:

- Place-based models forecast hotspots in small geographic grids or areas where specific crimes appear more probable. Inputs include historical incident data on type, location, and timestamp, along with environmental variables and contextual factors such as proximity to crowded areas, festivals, or local events.
- Person-based models assign risk scores to individuals for offending or victimization. These models often use network analysis of associations, criminal history, demographics, and behavioural trends.

Common algorithms include

- Regression techniques and time-series models for baseline forecasting.
- Tree-based ensembles such as Random Forest and XGBoost handle mixed data types and provide feature importance.

- Deep learning methods address complex spatiotemporal patterns through convolutional or recurrent networks that process grid-based crime data.
- Risk terrain modelling integrates with machine learning, combining crime history with social or cultural events and local factors.

Performance evaluation relies on metrics such as precision, recall, F1-score, ROC-AUC, and hit rates that measure the percentage of predicted hotspots experiencing crime. Academic studies using Indian city data from sources such as the National Crime Records Bureau (NCRB) have reported high accuracy in models like Random Forest for crime rate forecasting across multiple cities, with R^2 scores reaching 0.92 in some cases. Real-world utility depends on operational integration rather than academic metrics alone. Many systems deliver modest improvements over traditional methods. Recent developments in India highlight integration of AI into national platforms for improved interpretability, although many tools remain under development or partially deployed.

Prominent Case Studies

1. Delhi Police Initiatives: CMAPS and Related Systems

The Delhi Police developed the Crime Mapping, Analytics and Predictive System (CMAPS) in collaboration with the Indian Space Research Organisation around 2015. This web-based application supports live spatial hotspot mapping, analysis of criminal behaviour patterns, and suspect identification for all police stations. It incorporates elements of predictive analytics for crime-prone areas, with extensions into facial recognition and image enhancement tools for investigations. Reports indicate use in resource allocation for theft, assault, and other offenses in high-density urban settings. Delhi has also integrated automated number plate recognition and AI-driven CCTV analytics under Safe City projects. Outcomes include faster case resolutions in some instances, yet evaluations note challenges in scaling across diverse neighborhoods.

2. Mumbai Police Predictive Policing System

Mumbai Police deployed an AI-powered predictive system focused on crimes such as mobile theft, chain snatching, and vehicle theft. Developed with academic collaboration, the model analyses over five years of historical data, incorporating location, time, weather, and event factors to generate weekly hotspot predictions and patrol recommendations. Pilot implementations achieved around 75 percent accuracy in forecasting specific crime types. The system shifts policing toward proactive deployment, enabling targeted patrols that reduce response times and prevent incidents. Mumbai has combined this with broader surveillance under Safe City initiatives, showing improvements in certain property crimes.

3. CCTNS 2.0 National Platform

The Crime and Criminal Tracking Network & Systems (CCTNS) connects over 17,000 police stations nationwide. Its second version, under development as of 2026, integrates artificial intelligence for predictive policing, criminal profiling, and risk scoring. Features include place-based predictions using Risk Terrain Modelling that accounts for crime history, local events, and factors such as drug hotspots. Person-based elements support suspect profiling and identification of repeat offenders. The Ministry of Home Affairs has outlined these capabilities to enable beat planning and cross-jurisdictional analysis. Full rollout remains ongoing, with expectations of enhanced hotspot forecasting and resource optimization across states.



4. Other State and City Examples B

Bengaluru Safe City Project employs AI video analytics and integrated surveillance for crime detection and hotspot identification, contributing to resolutions of complex cases through movement tracing. Kerala Police have adopted predictive approaches and used AI for aging suspect images in cold cases, leading to breakthroughs via social media cross-referencing.

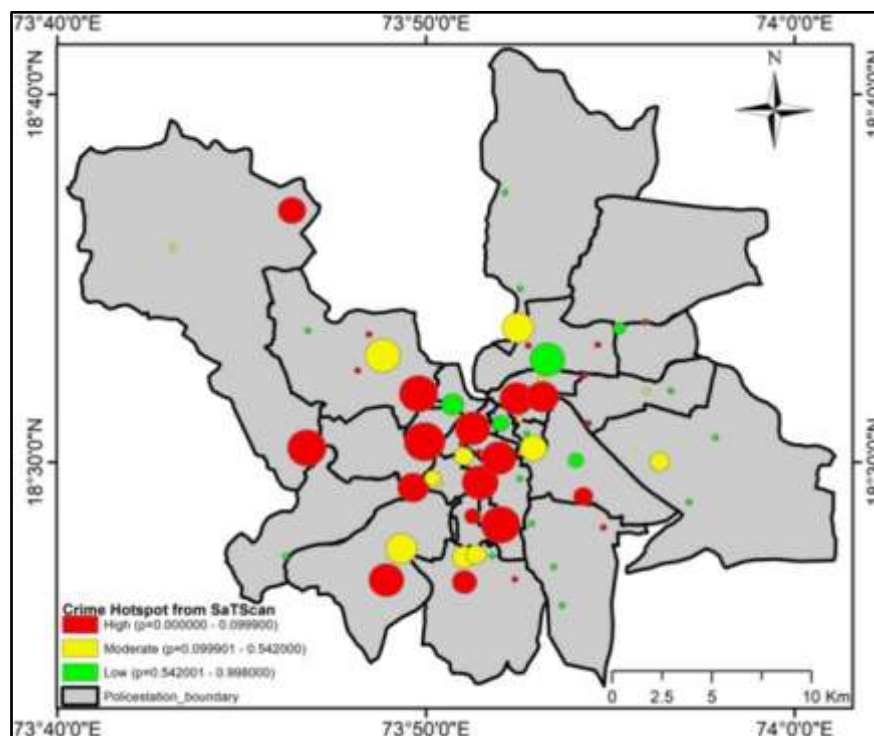
Maharashtra introduced MahaCrimeOS, an AI-powered operating system for cybercrime investigations, with broader implications for predictive tools. Telangana's TSCOP app and systems in Punjab, Rajasthan, and others incorporate facial recognition and analytics. Academic models for cities including Delhi, Mumbai, and Bengaluru demonstrate potential through geospatial analysis and machine learning on NCRB data spanning multiple crime categories.

International comparisons appear in research, yet Indian deployments emphasize integration with existing databases like CCTNS, e-Prisons, and e-Forensics.

Empirical Outcomes and Effectiveness

Evidence from Indian implementations remains varied. Mumbai pilots reported 75 percent accuracy for targeted property crimes, enabling better patrol allocation. Delhi systems aided investigations in hundreds of cases, including riot-related matters through facial recognition support. Academic studies on urban data from 19 cities achieved strong forecasting performance with low error metrics in regression tasks for crime rates.

Some departments note resource optimization and faster responses in high-risk zones. Broader reviews suggest modest gains in proactive measures when combined with traditional policing. However, many outcomes depend on data quality and integration levels. Certain analyses indicate limited long-term crime reduction without complementary social interventions. Place-based tools show more consistent results than person-based scoring in preliminary evaluations.



Crime hotspots heatmap of Pune, Maharashtra

Challenges: Bias, Ethics, and Limitations

Historical crime data in India often reflects policing patterns that have disproportionately affected certain communities, including marginalized groups and minority neighbourhoods. Algorithms trained on such data can perpetuate cycles in which over-policing leads to more records and heightened future predictions for those areas. Reports highlight risks of algorithmic

bias in facial recognition and hotspot mapping, where performance varies across demographics due to dataset imbalances.

Additional issues include opacity in some proprietary or developing algorithms that limit auditing. Officers may over-rely on outputs, which reduces discretionary judgment. Integration with extensive surveillance raises privacy concerns under frameworks such as the Digital Personal Data Protection Act. Feedback loops can reinforce existing disparities. Data quality varies across states, with incomplete records or inconsistencies undermining reliability. Parliamentary committees have flagged risks related to bias, surveillance overreach, and civil liberties in predictive tools.

Analyses describe predictive policing in India as a socio-technical challenge that requires oversight beyond technical deployment, especially given constitutional considerations around equality and due process.

Future Directions and Recommendations

Advances in explainable AI, privacy-preserving techniques, bias-mitigation methods, and multimodal data from non-police sources such as urban planning offer pathways forward. Hybrid models that combine AI outputs with community insights and rigorous evaluations hold value.

Government reports and guidelines emphasize pre-deployment assessments for fairness. Transparency measures and explainability features support accountability. Human oversight with defined policies helps prevent undue reliance. Regular audits and engagement with stakeholders build confidence. Programs can emphasize prevention by linking predictions to interventions such as social services or community programs. Regulatory approaches seek to balance innovation with rights protection.

In the 2025–2026 period, AI predictive tools in India continue expanding through national platforms like CCTNS 2.0 and state initiatives. Some efforts focus on limited-scope applications with attention to governance.

AI-based crime prediction offers data-driven insights that could optimize limited resources and support prevention in India's diverse urban and rural contexts. These systems also carry risks of embedding societal inequities without careful safeguards. Case studies from Delhi, Mumbai, Bengaluru, and national platforms illustrate operational potential alongside practical difficulties, particularly in bias management and equitable outcomes.

Success depends on governance through transparent design, equitable data practices, independent evaluation, and alignment with broader public safety goals. As a socio-technical system, AI crime prediction in India calls for input from data scientists, criminologists, legal experts, policymakers, and communities. Future research should emphasize causal studies, context-specific fairness metrics, and approaches that strengthen public trust. The technology contribution will ultimately reflect its support for effective and equitable policing across the country.